INCH-POUND
A-A-52435B(AT)
10 January 2025
SUPERSEDING
A-A-52435A
3 August 2009

COMMERCIAL ITEM DESCRIPTION

HOSE ASSEMBLIES, METAL: FLEXIBLE, FUEL AND OIL

The General Services Administration has authorized the use of this commercial item description (CID) as a replacement for MIL-H-13524B(AT), which is cancelled.

1. SCOPE. This CID covers two types of flexible hose assemblies for the conveyance of gasoline or oil or for other applications when installed in military vehicles or other U.S. Army equipment with the exception of use with aircraft engine fluid lines.

2. CLASSIFICATION. Hose assemblies shall be of the following types as specified (see 7.2):

2.1 Types.

Type I - Corrosion-resistant steel
Type II - Bronze or brass

3. SALIENT CHARACTERISTICS

3.1 Materials. Materials used in the hose assemblies shall be as specified herein and in applicable drawings (see 7.2). The use of recovered materials made in compliance with regulatory requirements is acceptable providing all requirements of this CID are met (see section 4).

3.2 Design and construction. Hose shall consist of corrugated or convoluted tubing of seamless, welded, or brazed design, protected with a cover of one or more beamed wire braids. Each length of hose shall be fitted with a set of couplings, as specified in applicable drawings (see 7.2), each permanently attached to hose.

Beneficial comments, recommendations, additions, deletions, clarifications, and any other data that may improve this document should be sent to U.S. Army DEVCOM Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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3.3 Dimensions. Hose assemblies shall be of the dimensions specified in procurement documents and in applicable drawings (see 7.2).

3.4 Hose assembly interchangeability. Unless otherwise specified, hose assemblies for each particular size (inside diameter) shall be interchangeable with all other like hose assemblies intended for the same use.

3.5 Performance.

3.5.1 Flexibility and leakage. Hose assemblies shall be capable of being bent around a mandrel under an internal hydrostatic pressure in accordance with table I while at 300 degrees Fahrenheit (°F) [149 degrees Celsius (°C)]. This shall be evidenced by the absence of leaks, deformation, permanent set, or interior obstruction.

TABLE I. Mandrel diameters and hydrostatic pressure for flexibility and leak test.

Hose size (inside diameter)		Mandrel diameter		Hydrostatic pressure	
inches	millimeters	inches	millimeters	pounds per square inch	Kilopascals
3/16	4.8	5-1/2	139.7	1830	12617
1/4	6.3	7	177.8	1470	10135
5/16	7.9	8	203.2	1160	7998
3/8	9.5	11	279.4	890	6136
1/2	12.7	14	355.6	820	5654
5/8	15.9	18	457.2	750	5171
3/4	19.0	20	508.0	660	4551
1	25.4	24	609.6	440	3034

3.6 Environmental.

3.6.1 Vibration. The hose assemblies shall show no evidence of leakage after being subjected to 500 hours of vibration, followed by 24 hours of refrigeration at -65°F and subsequent flexing through 90 degree bends in four directions, followed by an additional 24 hours of vibration. The vibration shall be performed with SAE 30 oil circulating at 300°F (149°C), 175 pounds per square inch (psi) [1207 Kilopascals (kPa)], 116°F (47°C) ambient air temperature, .052 inch [1.3 millimeters (mm)] total displacement (double amplitude), at 30 cycles per second. Prior to vibration, the sample shall be arranged to provide an initial slack of 3/16 inch (4.8 mm) and an offset of 1-1/2 inches (38.1 mm) at an angle of 90 degrees to the direction of stroke. The vibration may be performed continuously or intermittently.

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3.6.2 Corrosion resistance. When hose fittings are made of ferrous alloys, either in whole or part, they shall withstand 100 hours of salt spray in accordance with ASTM B117 without showing evidence of damaging corrosion.

3.7 Identification and markings. Identification and markings of the hose assemblies shall be permanent and legible and shall include as a minimum, the manufacturer's identification code (CAGE), the engineering drawing part number (see 7.2), and the national stock number (NSN).

3.8 Workmanship. Workmanship shall be such as to produce hose assemblies free from defects such as cracks, cuts, breaks, as well as restricted holes through end fittings, or damaged or burred end fitting threads.

4. REGULATORY REQUIREMENTS

4.1 Recovered, recycled, bio based, and environmentally preferable materials. The offeror/contractor is encouraged to use recovered, recycled, bio based, and environmentally preferable materials in accordance with Public Law 94-580 to the maximum extent practicable.

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

5.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

6. PACKAGING

Preservation, packing and marking shall be as specified in the contract or order (see 7.2).

7. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

7.1 Addresses for obtaining copies of referenced documents.

7.1.1 ASTM publications. Copies of ASTM documents may be obtained from www.astm.org.

7.1.2 IEEE publications. Copies of IEEE documents may be obtained from www.ieee.org.

7.1.3 Public law publications. Copies of Public Law 94-580 may be obtained from www.epa.gov.

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7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number and date of this commercial item description.
b. Issue of the ASSIST and industry standards to be cited in the solicitation (see 7.1).
c. Title, number, and revision letter of the applicable engineering drawing (see 3.2).
d. Type, size, and length of hose assemblies required (see 2.1, 3.1, and 3.3).
e. Engineering drawing part number (see 3.7).
f. Selection of applicable level of protection and packaging requirements (see section 6).

7.3 Cross-reference data. Hose assemblies conforming to this document are interchangeable/ substitutable with hose assemblies conforming to MIL-H-13524B(AT).

7.4 Metric product. Hose assemblies that are to metric dimensions will be considered on the following basis:

a. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of IEEE/ASTM SI 10, and all other requirements of this CID are met.
b. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.
c. The contracting officer has the option of accepting or rejecting the product.

7.5 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

MILITARY INTERESTS:

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

Custodian:
Army - AT

Preparing Activity:
Army - AT

Review Activities:
Army - EA
DLA - CC

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